The Islamia University of Bahawalpur
Department of Information Technology

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E-Commerce Application Development

Appendix

D

Glossary

Taught in —

Laboratory —

Assessment —

This is a class extract. It is *Appendix D* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

Dr. Mustafa Hameed · Department of Information Technology

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Chapter D

Glossary

Terms are defined as this course uses them. Where a word is used differently elsewhere in the industry, that is said. The chapter in brackets is where the term is introduced and where the evidence for any claim about it lives.

abandonment. A basket or checkout the customer began and did not finish. In this platform it is visible as the gap between rows in `oc_order` and orders above status 0. (Chapter 10, Chapter 17)

acquirer. The bank holding the shop's account, which accepts card payments on its behalf and carries the risk that the shop fails to deliver. (Chapter 13)

affiliate. A customer paid commission on orders arriving through their tracking code. Default commission here is 5%. (Chapter 12)

authorisation. A request asking the issuer whether a payment would succeed, reserving the amount. Reversible, instant, and not money. (Chapter 13)

bcrypt. A deliberately slow password hashing function with a tunable cost factor. PHP's `PASSWORD_DEFAULT` produces bcrypt at cost 10 here. (Chapter 15)

B2B, B2C, C2C. Business to business, to consumer, and consumer to consumer. A statement about who the parties are, not about the software. (Chapter 2)

capture. Claiming an amount previously authorised. Only now is the shop owed anything. (Chapter 13)

cash on delivery (COD). Payment collected by the courier on handover. Dominant in this market, and the most expensive instrument once refusals are counted. (Chapter 13)

chargeback. A card payment reversed by the issuer after a dispute. The loss falls on the shop. (Chapter 13)

container. A running instance of an image, with its own filesystem and process space, sharing the host's kernel. Not a virtual machine. (Chapter 5)

content security policy (CSP). A response header listing where a page may load scripts, styles and images from. Enforced by the browser. (Chapter 16)

cookie. A named value the server asks the browser to store and return on later requests. The only standard way to relate two requests. Lives on the customer's machine, where they can read and change it. (Chapter 9)

cost factor. Bcrypt's work parameter. Each step doubles the time for you and for an attacker. (Chapter 15)

coupon. A discount the customer must type a code to obtain. The code is what makes a campaign measurable. (Chapter 12)

credential stuffing. Trying email and password pairs stolen elsewhere. Defeats password rules, because the passwords are correct. (Chapter 15)

CSRF token. A one-time value in a form, checked on submission, so that another site cannot submit the form on a visitor's behalf. This shop's login carries one. (Chapter 15, Chapter 16)

destination principle. Tax belongs where the goods are consumed — hence a tax rule based on the shipping address. (Chapter 11)

Docker Compose. A file describing several containers, their networks and their volumes, brought up together by one command. (Chapter 5)

gateway (payment). The software company between the shop and the acquirer. The only one of the five parties you integrate with. (Chapter 13, Chapter 14)

geo zone. A named set of (country, zone) pairs that shipping methods and tax rules attach to. It has no meaning of its own. (Chapter 11)

hosted page. A payment architecture where the customer types their card on the provider's site. Keeps the shop in PCI DSS SAQ-A. (Chapter 14)

idempotency key. A value sent with a request so that repeating it has the effect of sending it once. Must belong to the order, not the request. (Chapter 14)

image (container). A read-only layered filesystem holding an application and its dependencies. Built once, identical everywhere. (Chapter 5)

issuer. The bank that gave the customer their card or holds their wallet. Decides whether to approve, and hears their complaints. (Chapter 13)

long tail. A latency distribution where a minority of requests take far longer than the median, so the mean sits well above it. A timeout must be chosen against the tail. (Chapter 14)

marketplace. A venue where third parties sell, with the operator taking a fee and holding neither stock nor the customer relationship. (Chapter 2)

merchant discount rate. The single percentage a shop pays on a card payment, divided afterwards between scheme, acquirer and gateway. (Chapter 13)

module. A unit of functionality added to the platform without editing it: a shipping method, a payment method, a dashboard panel. (Chapter 19)

option (product). A choice offered on a product — size, colour — stored as rows that grow by addition. Contrast *variant*. (Chapter 7)

order. A row recording a commitment to buy, with the products, totals and addresses *copied* as they were at the time. It does not point at the catalogue. (Chapter 10)

order history. The list of an order's status changes with dates. The status says where an order is; the history says how it got there, and is what a dispute is settled from. (Chapter 10, Chapter 18)

order status. The state an order is in. The set is the shop's own, not a standard. Status 0 is not in the set. (Chapter 10)

PayPak. Pakistan's domestic card scheme, running over the 1LINK switch. Cheaper to accept and domestic only. (Chapter 13)

PCI DSS. The card industry's data security standard. Its reporting burden is graded by how much card data you touch; a hosted page keeps you on the shortest questionnaire. (Chapter 14)

PECA 2016. Pakistan's Prevention of Electronic Crimes Act. Unauthorised access to an information system is an offence under it, with no educational exemption. (Chapter 16, Chapter 22)

percentile. The value below which a given share of measurements fall. The 95th percentile of a response time tells you what a slow customer experiences; the mean does not. (Chapter 14)

Raast. The State Bank's instant payment system, settling bank to bank in seconds at negligible cost. (Chapter 13)

return-to-origin. A delivery refused at the door and sent back. The shop pays both legs and keeps the goods. (Chapter 13)

salt. A random per-user value mixed into a password hash, so that identical passwords hash differently and precomputed tables are useless. Need not be secret. (Chapter 15)

session. Server-side state filed under a key, with the browser carrying only the key. Distinct from the cookie that points at it. (Chapter 9)

session engine. Where session records are kept — database, files or Redis. It does not decide where the basket is. (Chapter 9)

settlement. The acquirer actually paying the shop, days later and net of fees. (Chapter 13)

sort order (totals). The position of a total line in the pipeline. It decides what each line is computed from, and therefore what the customer pays. (Chapter 12)

special. A dated replacement price for a product and customer group, applied automatically. (Chapter 12)

tax base. Which address a tax rule is decided by: store, payment or shipping. The most consequential dropdown in the admin panel. (Chapter 11)

tax class. A kind of product for tax purposes, such as taxable goods or downloadable products. (Chapter 11)

tax rate. A number and the geo zone it applies in. Percentage or fixed amount. (Chapter 11)

tax rule. The join between a class and a rate, with a priority and a base. (Chapter 11)

template override. A copy of one theme file in a child theme, replacing the original for that one page. The override directory is the list of your changes. (Chapter 8)

theme. A directory of Twig templates and assets. Never edit the default one. (Chapter 8)

timing side channel. Information leaked by how long an operation takes. Here, a registered email answered 65 ms slower than an unknown one. (Chapter 15)

Twig. The template language this platform renders pages with. It auto-escapes output, which is why the script probes in Chapter 16 did nothing.

user enumeration. Discovering which accounts exist, by timing, messages or behaviour. Reconnaissance rather than an attack, and worth closing for that reason. (Chapter 15)

variant. A product generated per combination of choices, growing by multiplication. Contrast *option*. (Chapter 7)

volume (container). Storage managed by the container runtime and outliving any container. `down` keeps it; `down -v` deletes it. (Chapter 5, Chapter 9)

voucher (gift). A stored-value instrument bought by one person and spent by another. HEC's outline names it; **this platform does not have one**. (Chapter 12)